Spam Detection Analysis

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Abstract

The study in question concerns the analysis of a logistic regression model, which belongs to the family of GLM models. In particular, we will analyze, through some variables, the possibility that a received email may be Spam or not. By using appropriate statistical techniques, we will see which variables will actually be significant for the analysis. Finally, we will use a dataset to make predictions, in order to see how effective the applied model has turned out to be.

Let's see, first of all, a brief introduction regarding the variables.

1. Introduction

The problem

After mentioning the problem in the abstract, let's see what the main characteristics of the used data are. The dataset, available on www.kaggle.com, contains 20,000 different e-mail and 6 variables.
The following are the last ones:

- num_links: a counting variable (0 to 9) that shows the number of links present in the email body. More links often mean higher chances of spam.
- num_words: total number of words in the email randomly picked between 20 and 200.
- has_offer: binary variable (0 or 1); 1 if the email contains the word "offer". Marketing language like "offer" is common in spam.
- sender_score: a simulated reputation score of the email sender (Float between 0 and 1). A low sender score means the sender is less trustworthy.
- all_caps: whether the subject line is written in ALL CAPS (binary 0-1). All-caps subject lines are usually attention-grabbing and common in spam.
- is_spam: the target variable, a binary 0-1, so distributed: 1: the mail is spam 0: the mail is not spam

Exploratory Data Analysis

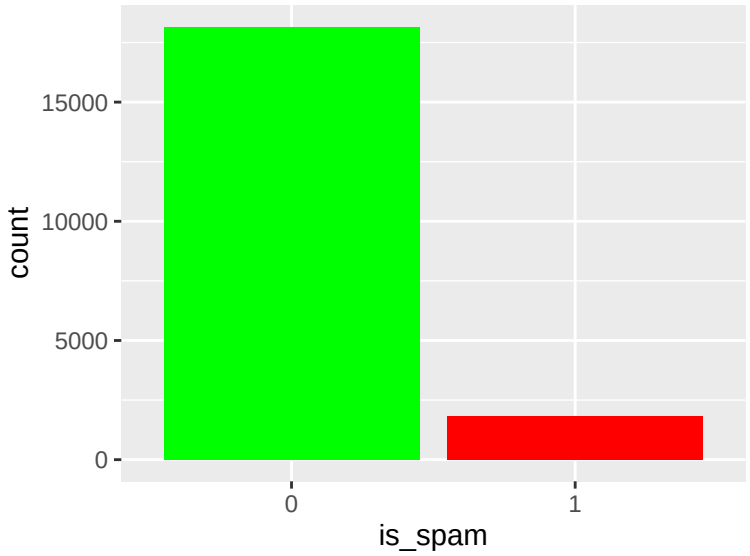
We start with a summary of the data to gain a general understanding of them structure, including the distributions of features, data types, and potential anomalies.

```
##    num_links       num_words     has_offer  sender_score    all_caps  is_spam
##  Min.   :0.000   Min.   : 20.0   0:13945   Min.   :0.0000   0:18044   0:18165
##  1st Qu.:1.000   1st Qu.: 64.0   1: 6055   1st Qu.:0.5671   1: 1956   1: 1835
##  Median :1.000   Median :110.0             Median :0.6997
##  Mean   :1.497   Mean   :109.5             Mean   :0.6942
##  3rd Qu.:2.000   3rd Qu.:154.0             3rd Qu.:0.8341
##  Max.   :9.000   Max.   :199.0             Max.   :1.0000
```

The summary provides an overview of the dataset's variables and their key statistics:

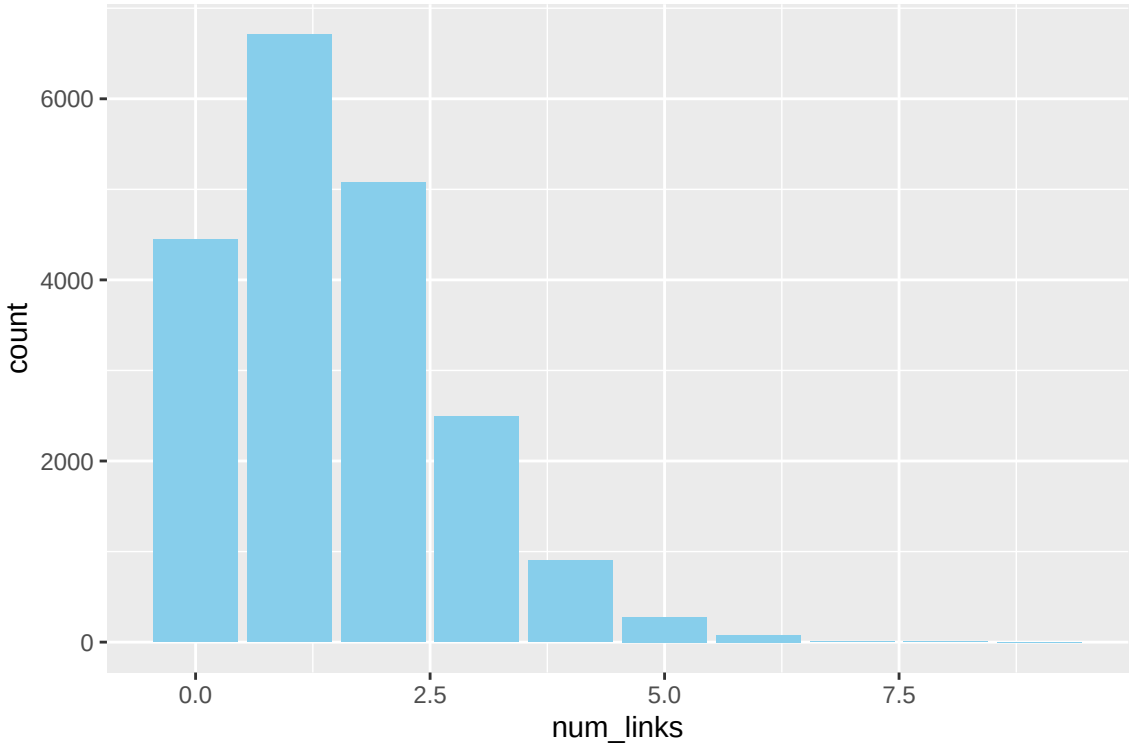
- num_links: we have a count from 0 to 9 of number of links. Median is equal to 1 and in mean there are almost 1.5 links per mail. This shows that we have few links in most observations and so could highlight the presence of few emails registered as potential spam.
- num_words: an average of about 110 words are recorded per email. It seems that there is a certain symmetry regarding that variable.
- has_offer: this variable shows there are many more e-mails that do not contain the word offer. In terms of frequency, we are talking about about 70% of the observations.
- sender_score: the mean and median values coincide and are both in the right part of the distribution; this means that most of the recorded score values are quite high (another value that highlights the presence of few spam emails).
- all_caps: only 10% of mails have the subject line written in ALL CAPS.
- is_spam: it represents the target variable, that's, the outcome within the logit model. From the summary, we note that only a minority of the observations (about 9%) is classified as spam. This could prove to be an important feature regarding forecasting; in fact, having such a high imbalance in the two classes may mean that: the threshold for class assignment, once posterior values are found, needs to be modified (the usual threshold of 0.5 may not generate optimal predictions).

Now, let's look at the characteristics of the variables graphically, starting from the target variable. We will then see the other variables, the interaction between the target and the others, and the interaction between some of them.

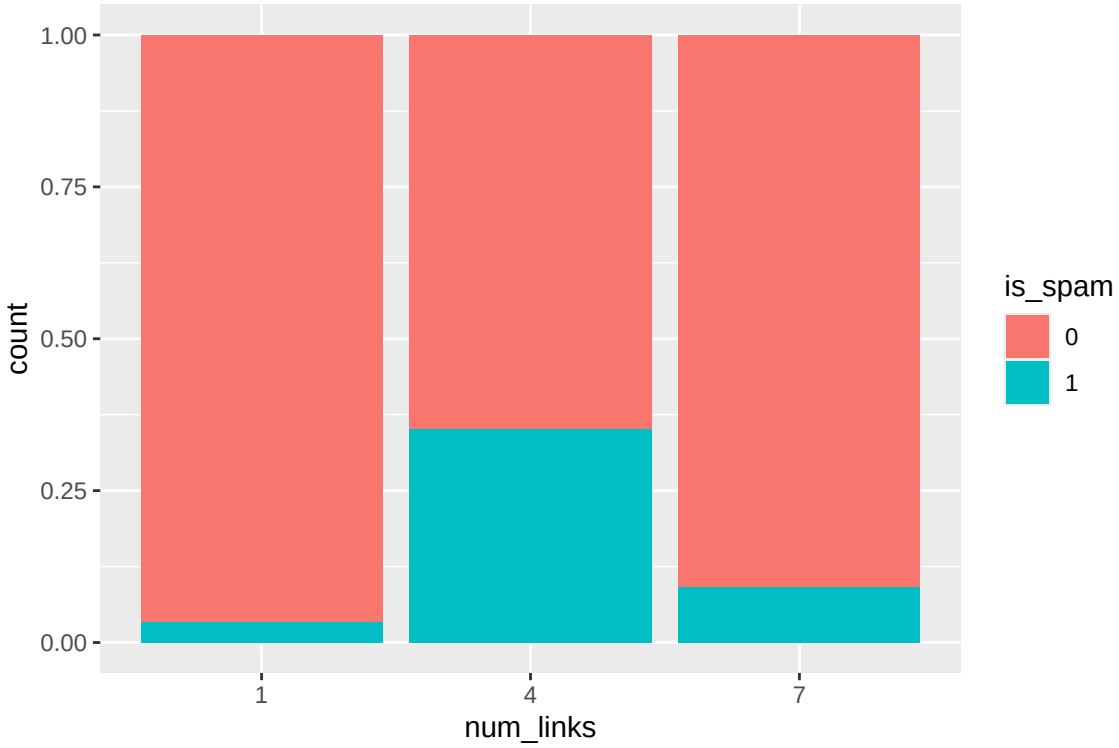


The bar plot shows the two classes in comparison. As we had already seen from the summary, the percentage of emails classified as spam is highly low.

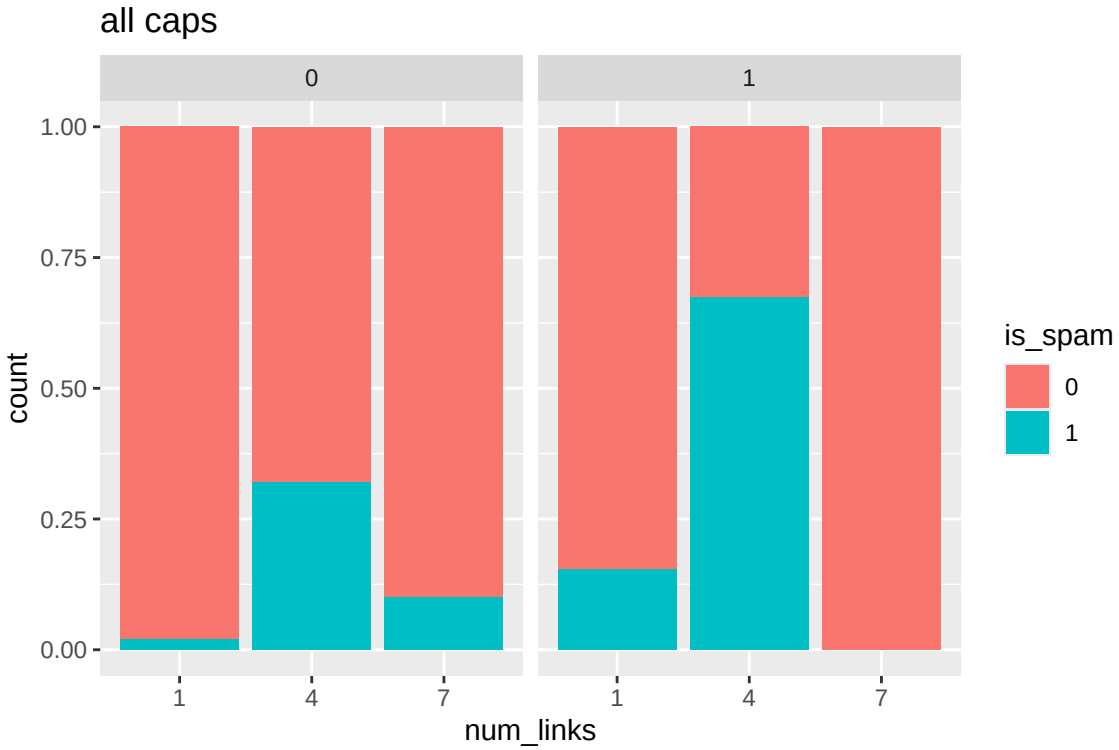
Let's see something about the number of links:

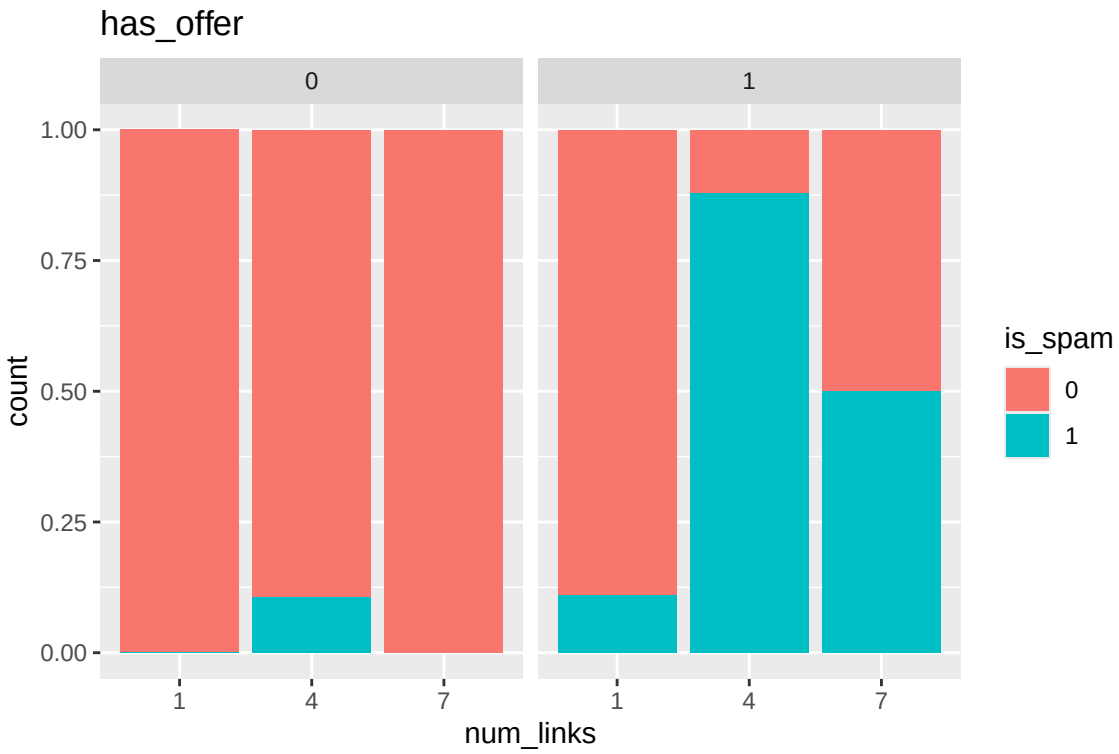


The frequency of number of links from 0 to 3 is significantly higher than the observations with a greater number of links. Given the number of levels represented, to have clearer conditional analyses we will select only 3 of the levels of the variable.



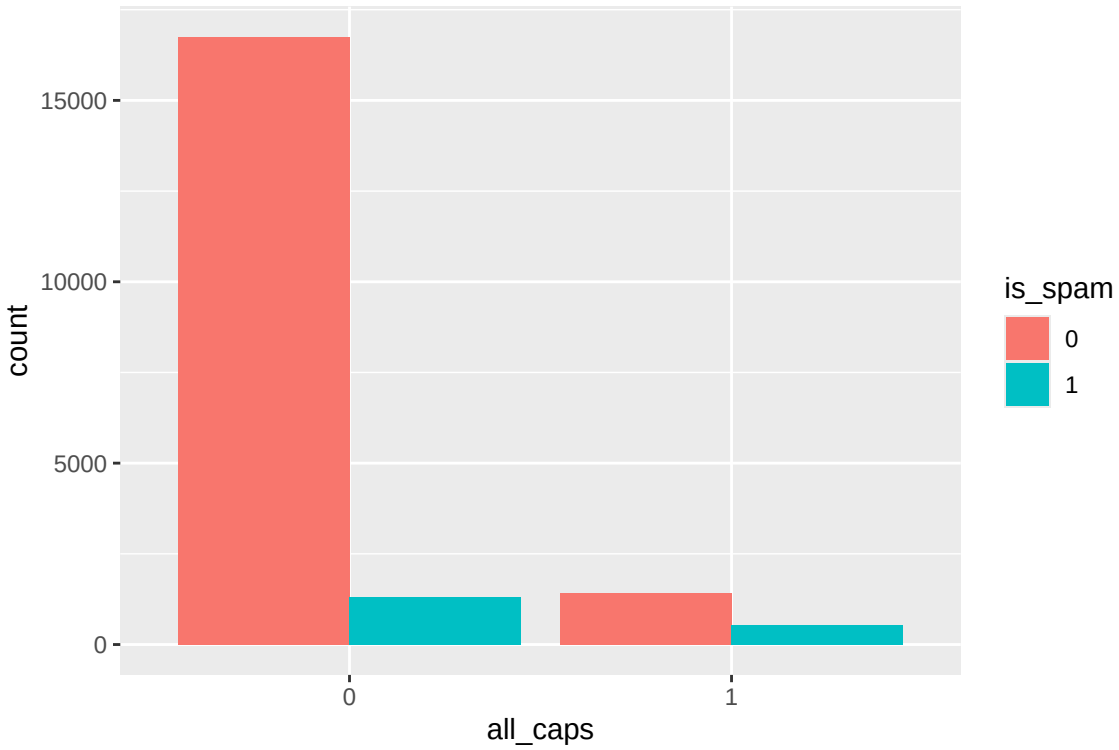
The graph presented above shows that the percentage of emails classified as spam increases significantly when going from 1 to 4 links, while when there are 7 links, the frequency tends to decrease, although still remaining higher than the value of 1. This could suggest that an average number of links between 0 and 9 has the highest percentage of spam.

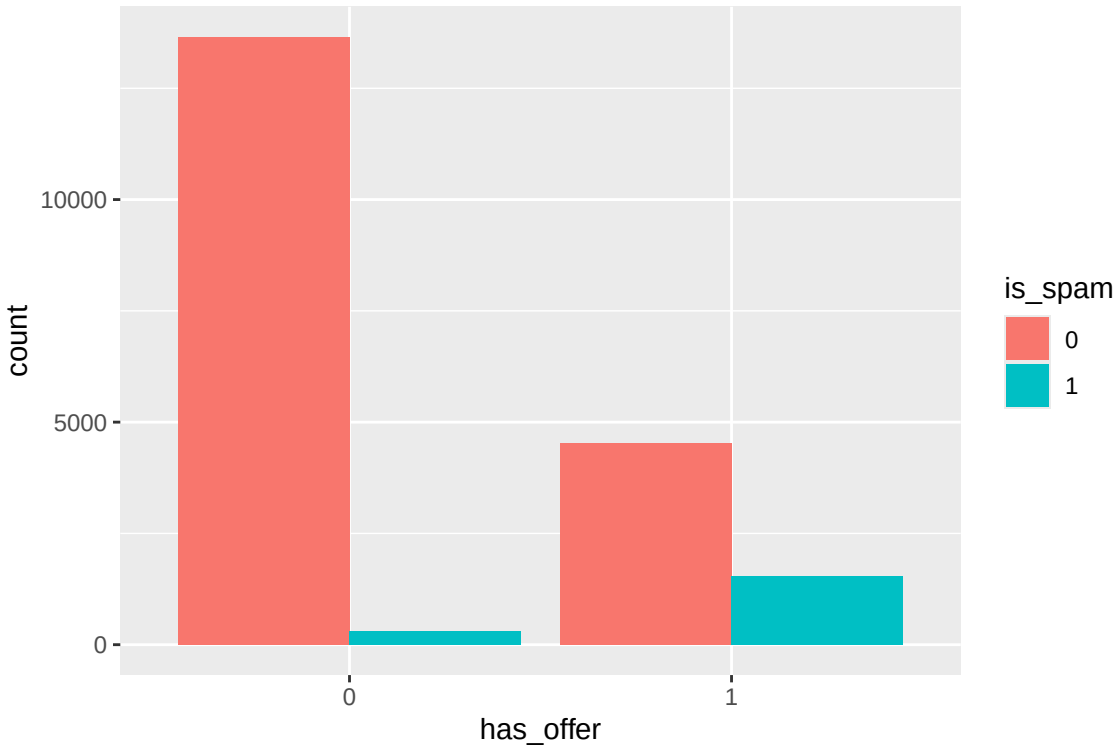




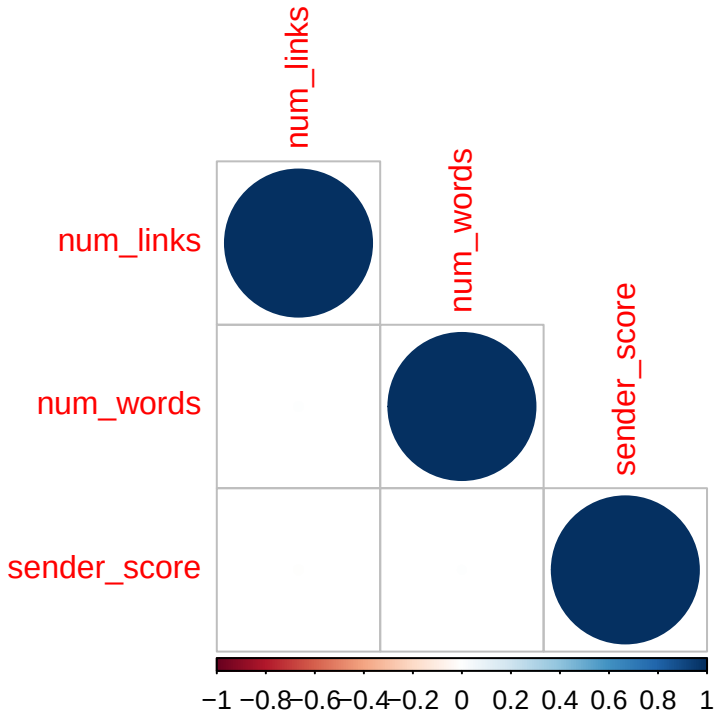
By further conditioning the previous graph on the two binary variables, we once again confirm what had already emerged from the summary:

- the word offer increases the percentage of spam.
- even have the subject line written in ALL CAPS produces the same effect.

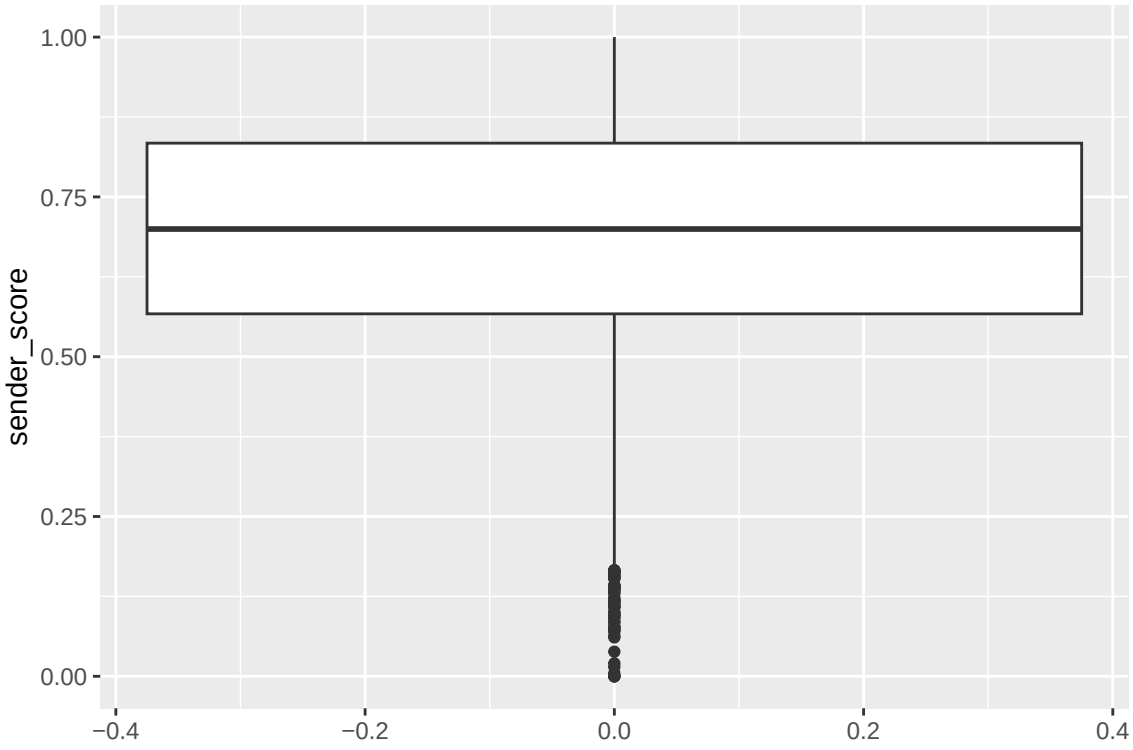


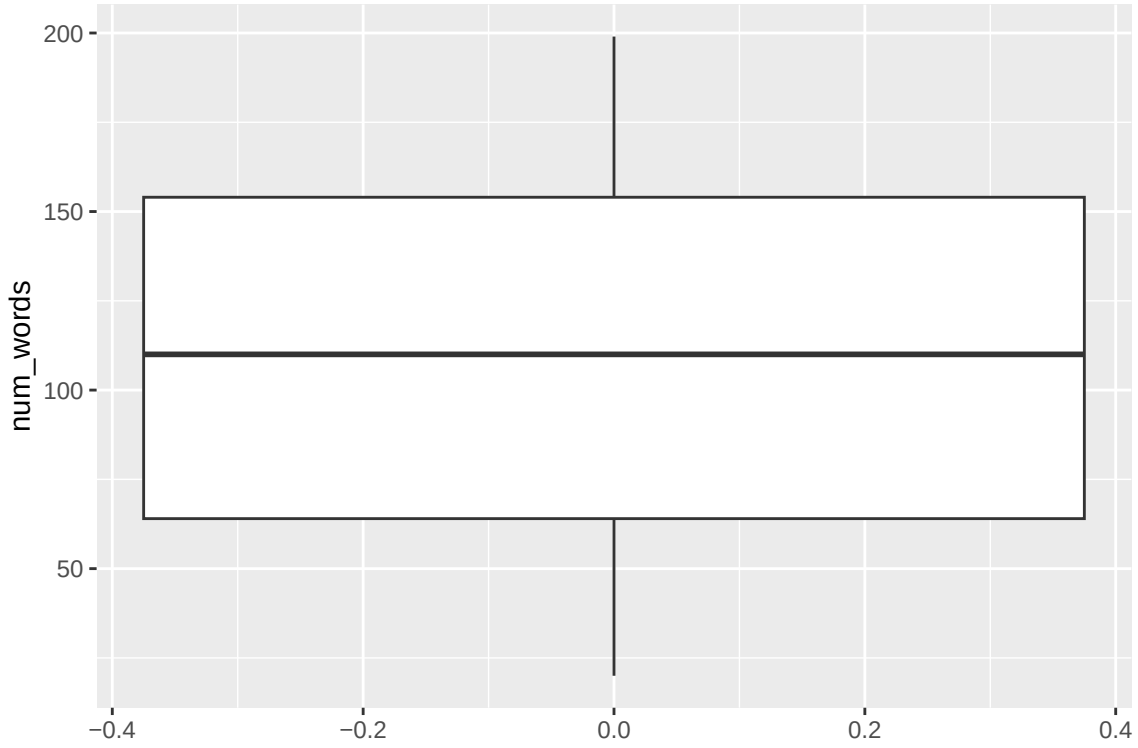


Let's now take a look at the only non-binary variables.



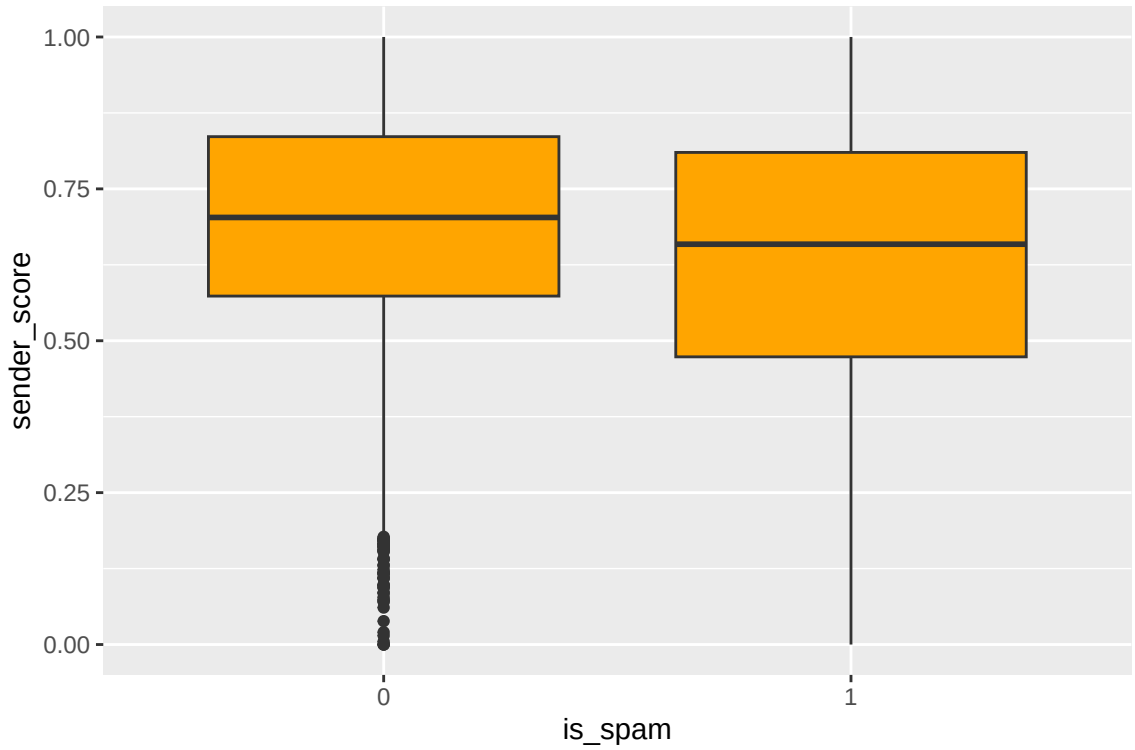
As we can see from the correlation matrix, there is no linear relationship between the variables. We therefore observe the variables individually.



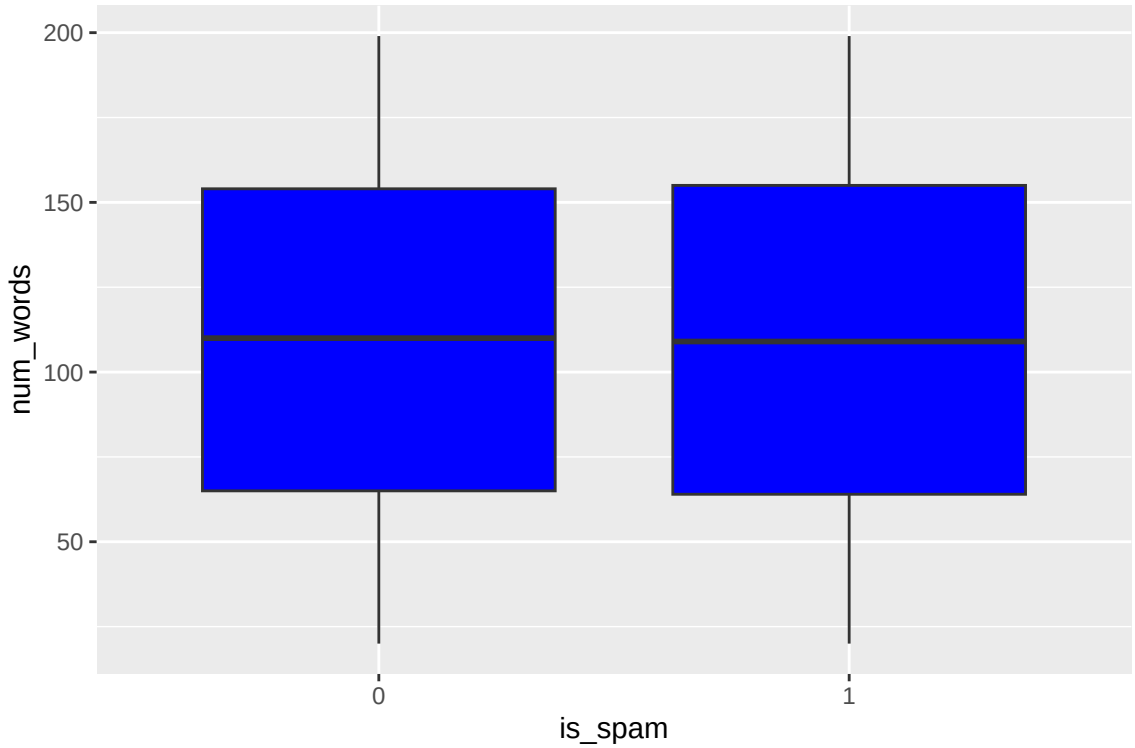


The distribution of the number of words is strongly balanced, while for the distribution of the variable related to the score we can confirm that most of the observations record a high score (more or less between 60 and 80%).

Let's see the interaction between these variables and the target variable.



We notice that when an email is marked as spam, the sender-score value slightly decreases. This matches what was stated at the beginning (a low sender score means the sender is less trustworthy).



There seems to be no difference in the number of words for spam emails and non-spam emails. Probably, the variable related to the number of words does not affect the classification of the observations.

2. Model estimation

The model

Now, we focus on the application of a logistic regression model to predict how the object should be classified. The logistic regression is particularly suited for cases of this type, that's when the outcome follows a bernoulli distribution (yes or no, good or bad...).

$$logit(\mu) = log(\frac{\mu}{1-\mu}) \quad con \quad Y|x \sim Bernoulli(\mu)$$

Where

$$\mu = \frac{exp(x'\beta)}{1 + exp(x'\beta)}$$

and so:

$$logit(\mu) = x'\beta$$

To ensure robust evaluation and reliable results, the dataset was divided into two subsets:

- Training set (80%): Used to train the model and identify the best predictors of the target variable.
- Test set (20%): Reserved for evaluating the model's predictive performance on unseen data.

Through a systematic model selection process, we identified the best-performing model, in which the variable num_words was excluded because it was not considered statistically significant.

```
##
## Call:
## glm(formula = is_spam ~ . - num_words, family = binomial(link = "logit"),
##     data = train_s)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -6.13551    0.18699  -32.81   <2e-16 ***
## num_links     1.34614    0.03664   36.74   <2e-16 ***
## has_offer1    4.28834    0.11332   37.84   <2e-16 ***
## sender_score -3.03476    0.20381  -14.89   <2e-16 ***
## all_caps1     3.14914    0.10912   28.86   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9828.5  on 15999  degrees of freedom
## Residual deviance: 4847.7  on 15995  degrees of freedom
## AIC: 4857.7
##
## Number of Fisher Scoring iterations: 7
```

The model can thus be written as:

$$logit(\hat{isSpam}) = -6.13 + 1.34 * numLinks + 4.28 * hasOffer1 - 3.03 * senderScore + 3.15 * allCaps1$$

Let's look at some information regarding the interpretation of the model. Since it is a logit link model, we cannot observe the actual magnitude from these estimates. The only thing we can say is whether the predictor has a positive or negative impact on the logarithm of the expected value of the is_spam variable.

Let's say that:

First of all, we can say that all the variables are statistically significant at the set α level.

- only the variable sender_score has a negative impact on the log-odds. This means that as the score increases, the probability of belonging to the spam class decreases; by how much? The Odds-ratio will tell us.
- num_links has a positive impact on the response variable. This obviously means the exact opposite of what was just explained. (increase in the probability of belonging to class 1 as the number of links present increases).
- has_offer and all_caps are both binary variables and have a positive impact on the log-odds. This means that the probability of belonging to the spam class is higher when the value of both is 1 compared to when they are both 0.

Other interpretations related to the estimated model concern:

- Null deviance: 9828 (degrees of freedom = 15999), which represents the deviance of a model with no predictors (null model).
- Residual deviance: 4847 (degrees of freedom = 15995), indicating a substantial reduction in deviance with the inclusion of predictors. This suggests the model explains much of the variability in the data.
- AIC: 4857: A lower AIC value indicates better model fit, though it's best compared with other candidate models. Among the models compared, this one has the lowest AIC value.
- The estimated dispersion parameter ϕ for the logistic regression is 1.

Interpretation of the coefficients

Let's see the actual magnitude of the estimated coefficients; to do this, we will calculate the odds-ratio of these.

$$\frac{\hat{\mu}}{1-\hat{\mu}} = exp(x'\hat{\beta})$$

Alternatively, with the expression $(exp(\beta_j) - 1) * 100$ we can describe the percentage impact that the predictor has on the expected value of Y (is_spam).

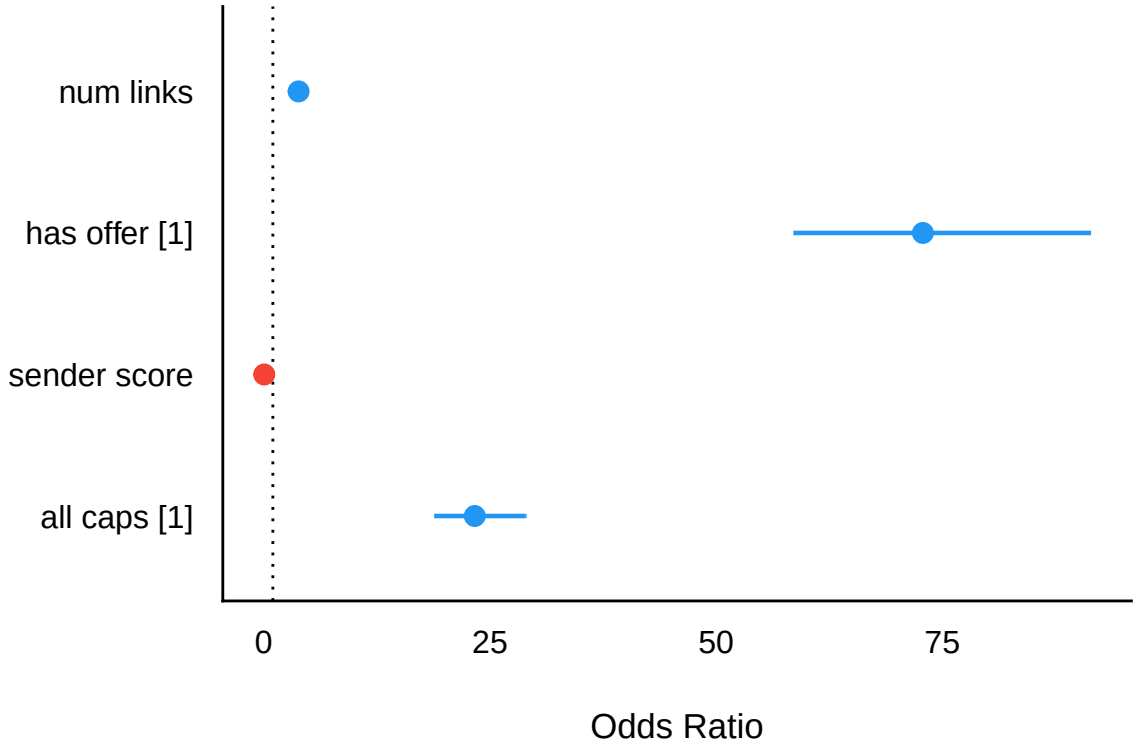
```
##  (Intercept)    num_links   has_offer1 sender_score    all_caps1
##    -99.78354    284.25517   7184.53557    -95.19138   2231.59300
```

- Intercept (-99.78): Represents the baseline expected value of the outcome when all predictors are at their reference or minimum values. In practical terms, this value serves as a starting point for the model.

- num_links (284.25): An increase in the number of links increases the probability of receiving a spam email by 284%.
- has_offer1 (7184.53): This is the variable that records the strongest impact within the model. In fact, having the word offer increases the likelihood of receiving spam by about 7000%.
- sender_score (-95.2): A unit increase in the sender's score reduces the probability of having a subject classified as spam by 95%. It is the predictor with the least recorded impact.
- all_caps1: When the subject line is written in ALL CAPS the probability that the object belongs to the spam category grows by 2200%.

We can therefore conclude that all the coefficients have a significant impact on the response variable and that has_offer1 is the variable that most affects it.

let's take a graphical look at what was just said.



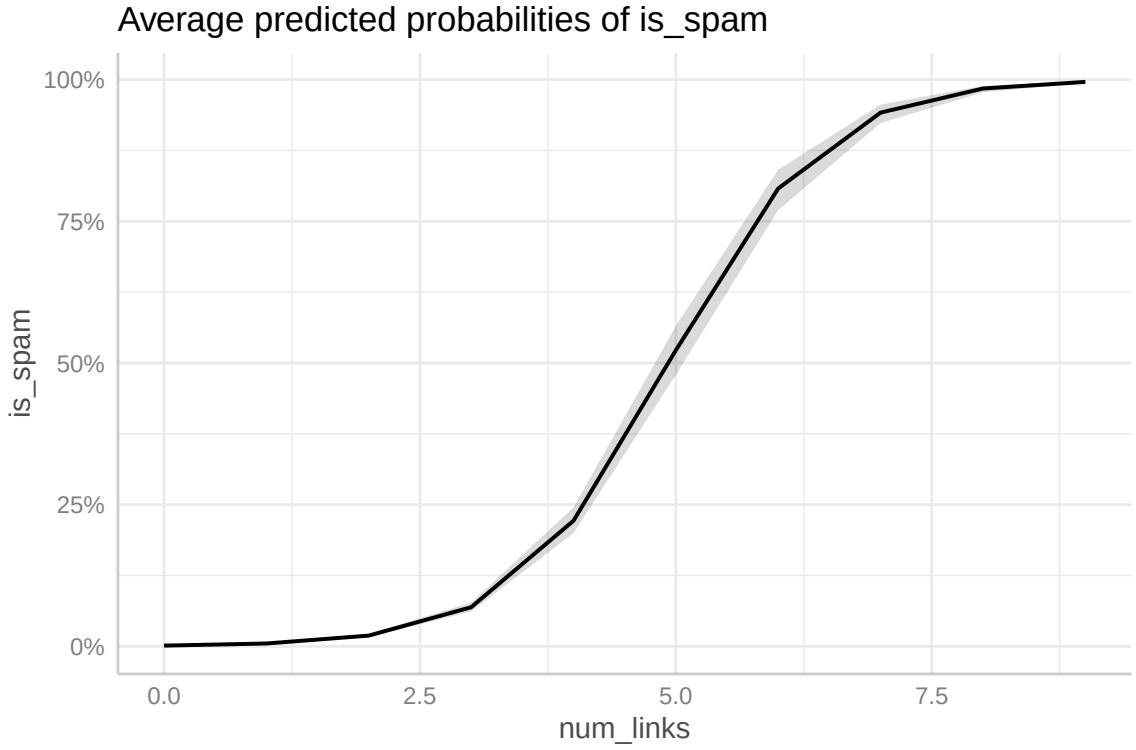
The odds ratio plot confirms the results obtained earlier. However, we notice high variability in the estimates of all_caps and especially of the variable has_offer, thanks to the confidence intervals.

Marginal effects

Let's take a look at the marginal effects.

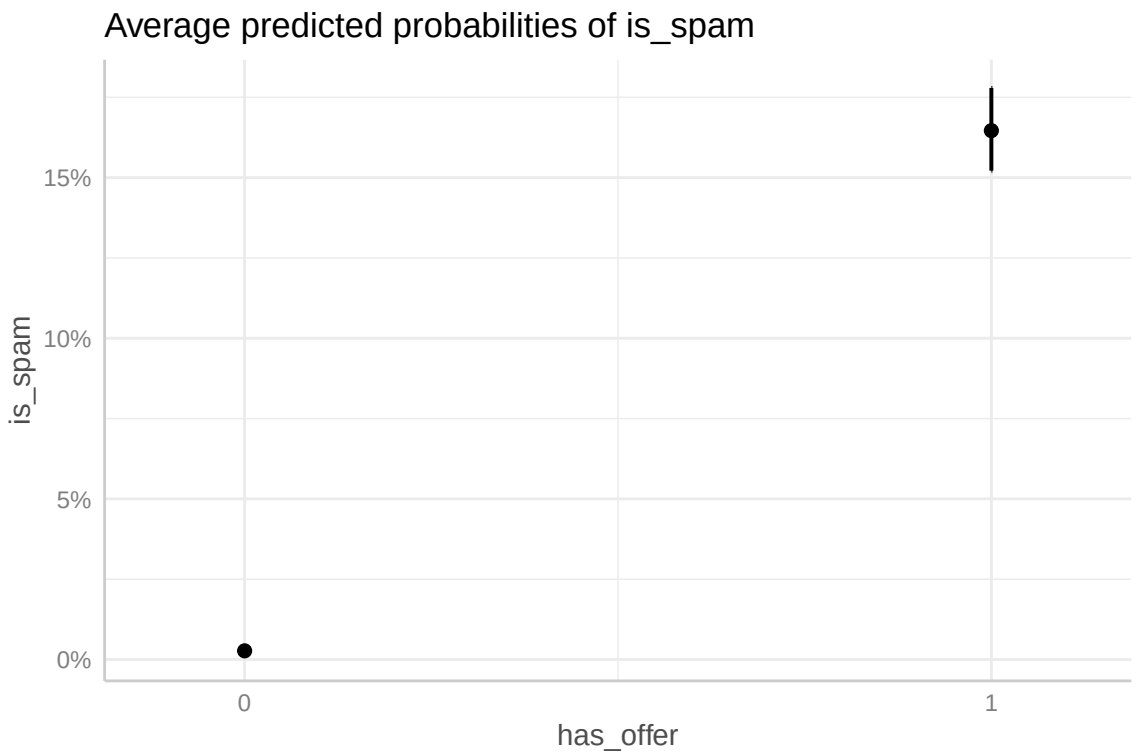
In a logit model, the marginal effects measure the change in the probability of an event following a one-unit change in an explanatory variable. They provide a more direct and intuitive response compared to the coefficient.

```
## # Average predicted probabilities of is_spam
##
## num_links | Predicted |     95% CI
## ----------------------------------
##         0 |      0.00 | 0.00, 0.00
##         1 |      0.00 | 0.00, 0.01
##         2 |      0.02 | 0.02, 0.02
##         3 |      0.07 | 0.06, 0.08
##         5 |      0.52 | 0.48, 0.57
##         6 |      0.81 | 0.77, 0.84
##         7 |      0.94 | 0.92, 0.96
##         9 |      1.00 | 0.99, 1.00
```



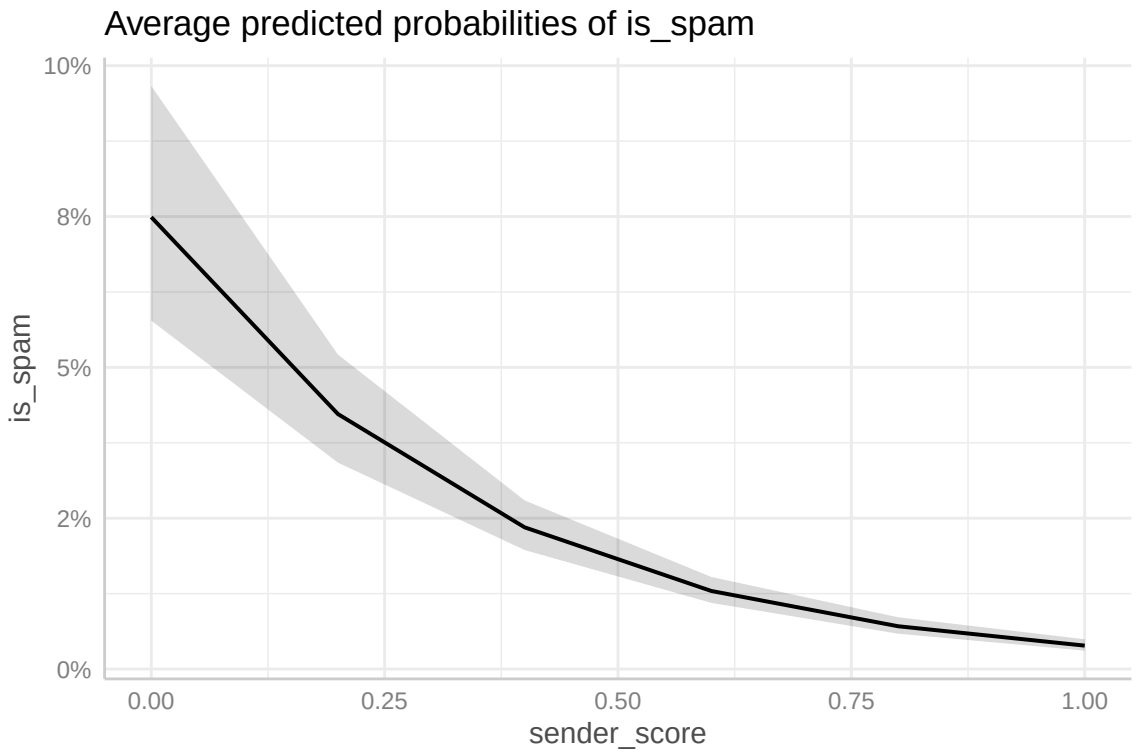
The graph shows a non-linear upward trend in is_spam with increasing num_links, indicating that the effect of the outcome becomes stronger as number of links increases. Confidence intervals are relatively narrow, suggesting high precision in these estimates.

```
## # Average predicted probabilities of is_spam
##
## has_offer | Predicted |     95% CI
## ----------------------------------
## 0         |      0.00 | 0.00, 0.00
## 1         |      0.16 | 0.15, 0.18
```



The plot highlights the strong impact that has_offer1 has on the probability that the object is classified as spam.

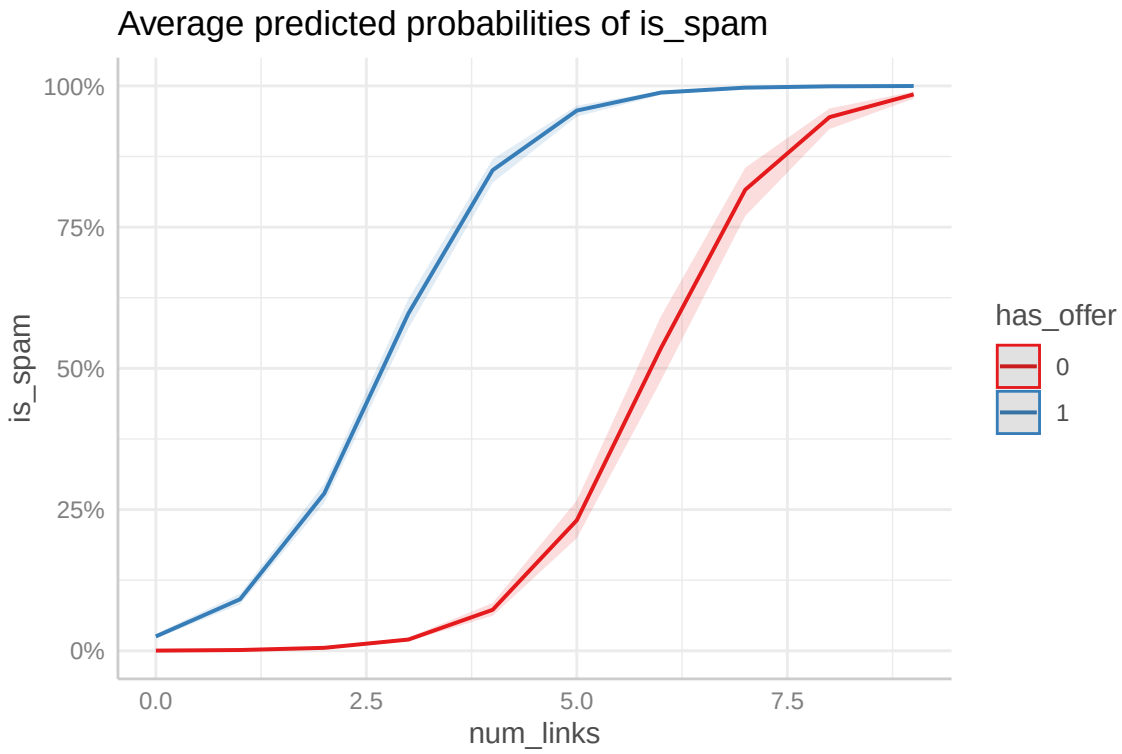
```
## # Average predicted probabilities of is_spam
##
## sender_score | Predicted |     95% CI
## -------------------------------------
##         0.00 |      0.07 | 0.06, 0.10
##         0.20 |      0.04 | 0.03, 0.05
##         0.40 |      0.02 | 0.02, 0.03
##         0.60 |      0.01 | 0.01, 0.02
##         0.80 |      0.01 | 0.01, 0.01
##         1.00 |      0.00 | 0.00, 0.00
```



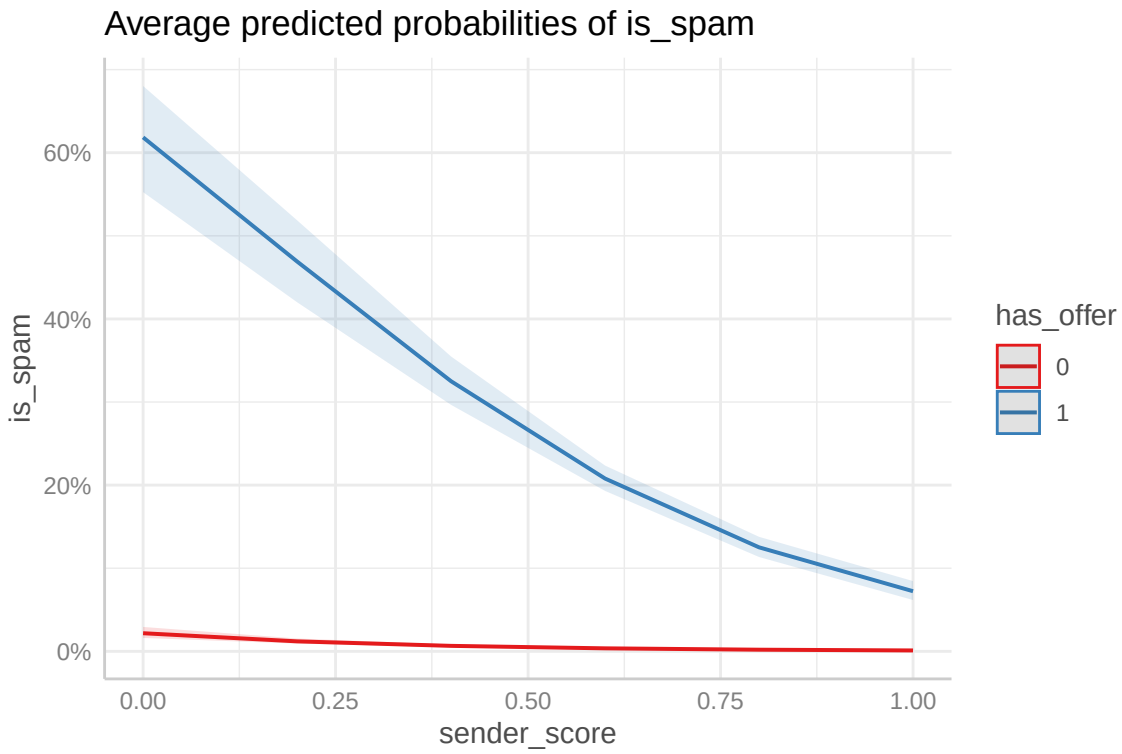
The value of is_spam decreases as sender_score increases. Effects that in this case become increasingly precise, taking into account the prediction range that tends to decrease as the score value increases.

Let'see some conditional marginal effect:

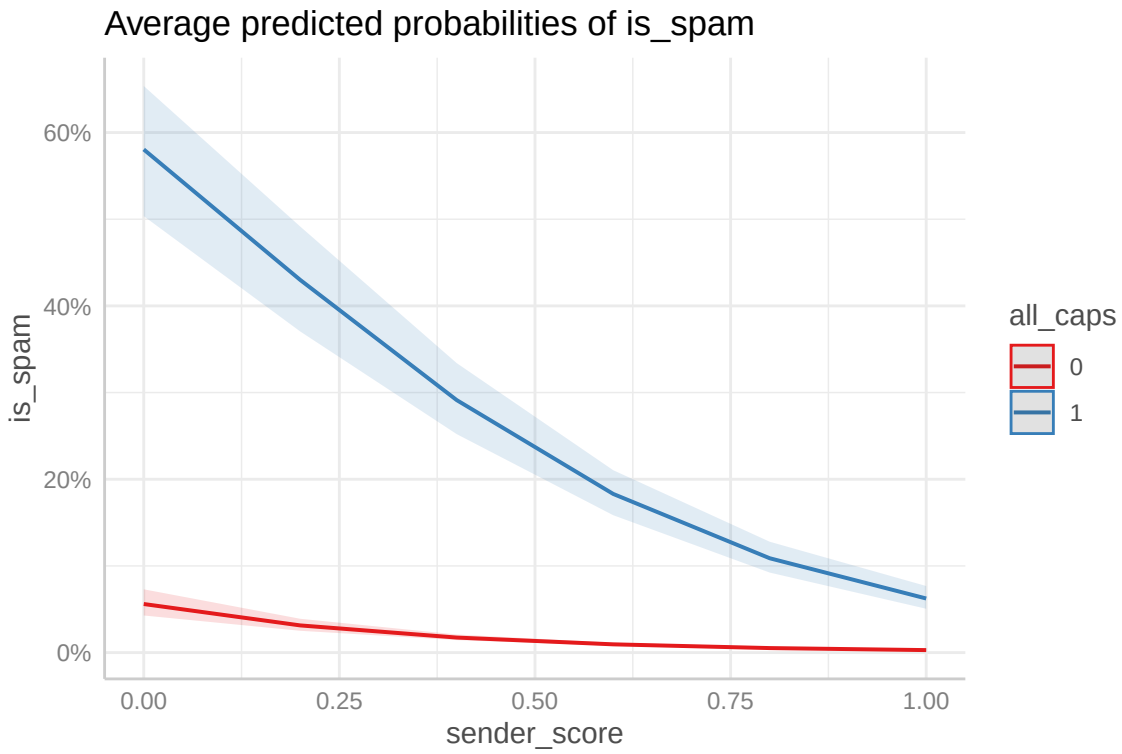
```
## # Average predicted probabilities of is_spam
##
## has_offer: 0
##
## num_links | Predicted |     95% CI
## ----------------------------------
##         0 |      0.00 | 0.00, 0.00
##         1 |      0.00 | 0.00, 0.00
##         3 |      0.02 | 0.02, 0.02
##         5 |      0.23 | 0.20, 0.27
##         6 |      0.54 | 0.48, 0.59
##         9 |      0.98 | 0.98, 0.99
##
## has_offer: 1
##
## num_links | Predicted |     95% CI
## ----------------------------------
##         0 |      0.03 | 0.02, 0.03
##         1 |      0.09 | 0.08, 0.10
##         3 |      0.60 | 0.57, 0.62
##         5 |      0.96 | 0.95, 0.96
##         6 |      0.99 | 0.98, 0.99
##         9 |      1.00 | 1.00, 1.00
```



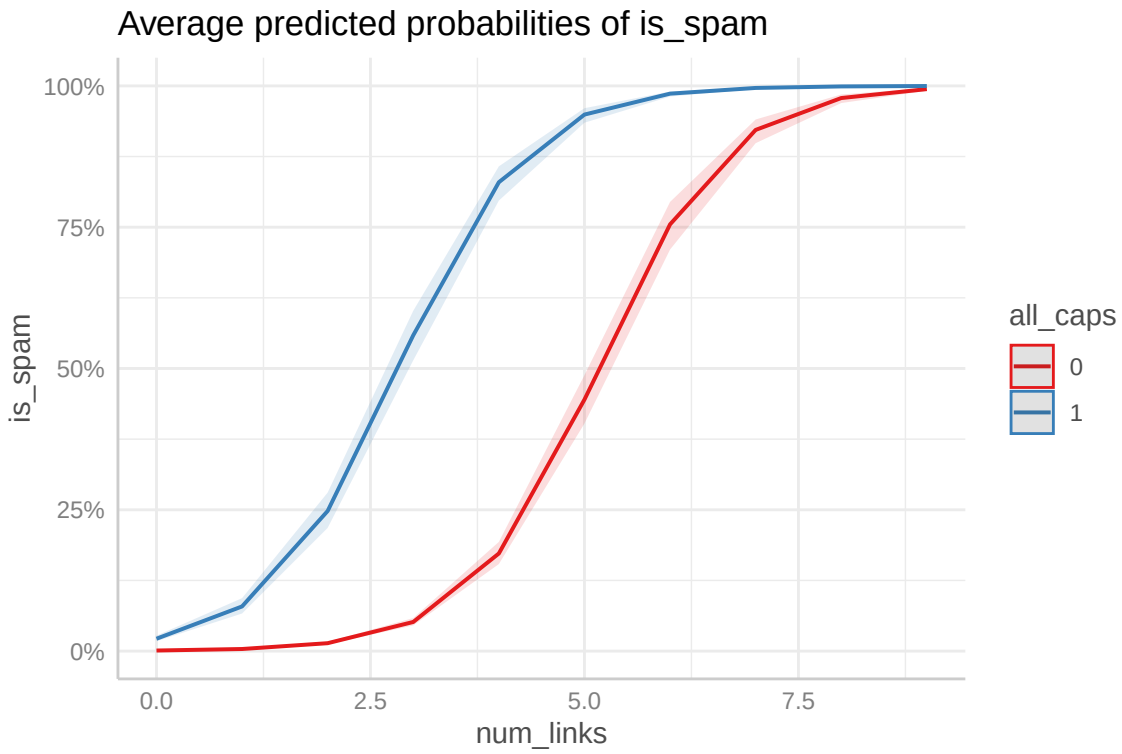

```
## # Average predicted probabilities of is_spam
##
## has_offer: 0
##
## sender_score | Predicted |     95% CI
## -------------------------------------
##         0.00 |      0.02 | 0.02, 0.03
##         0.20 |      0.01 | 0.01, 0.02
##         0.40 |      0.01 | 0.01, 0.01
##         0.60 |      0.00 | 0.00, 0.00
##         0.80 |      0.00 | 0.00, 0.00
##         1.00 |      0.00 | 0.00, 0.00
##
## has_offer: 1
##
## sender_score | Predicted |     95% CI
## -------------------------------------
##         0.00 |      0.62 | 0.55, 0.68
##         0.20 |      0.47 | 0.42, 0.52
##         0.40 |      0.33 | 0.30, 0.35
##         0.60 |      0.21 | 0.19, 0.22
##         0.80 |      0.13 | 0.11, 0.14
##         1.00 |      0.07 | 0.06, 0.08
```




```
## # Average predicted probabilities of is_spam
##
## all_caps: 0
##
## sender_score | Predicted |     95% CI
## -------------------------------------
##         0.00 |      0.06 | 0.04, 0.07
##         0.20 |      0.03 | 0.03, 0.04
##         0.40 |      0.02 | 0.01, 0.02
##         0.60 |      0.01 | 0.01, 0.01
##         0.80 |      0.01 | 0.00, 0.01
##         1.00 |      0.00 | 0.00, 0.00
##
## all_caps: 1
##
## sender_score | Predicted |     95% CI
## -------------------------------------
##         0.00 |      0.58 | 0.50, 0.65
##         0.20 |      0.43 | 0.37, 0.49
##         0.40 |      0.29 | 0.25, 0.33
##         0.60 |      0.18 | 0.16, 0.21
##         0.80 |      0.11 | 0.09, 0.13
##         1.00 |      0.06 | 0.05, 0.08
```




```
## # Average predicted probabilities of is_spam
##
## all_caps: 0
##
## num_links | Predicted |     95% CI
## ----------------------------------
##         0 |      0.00 | 0.00, 0.00
##         1 |      0.00 | 0.00, 0.00
##         3 |      0.05 | 0.05, 0.06
##         5 |      0.45 | 0.40, 0.49
##         6 |      0.75 | 0.71, 0.79
##         9 |      0.99 | 0.99, 1.00
##
## all_caps: 1
##
## num_links | Predicted |     95% CI
## ----------------------------------
##         0 |      0.02 | 0.02, 0.03
##         1 |      0.08 | 0.07, 0.09
##         3 |      0.56 | 0.51, 0.60
##         5 |      0.95 | 0.93, 0.96
##         6 |      0.99 | 0.98, 0.99
##         9 |      1.00 | 1.00, 1.00
```



The effects once again reflect the results obtained previously:

- the number of links increases the probability of having spam; probability that increases more quickly to 1 when the word offer is present and when the subject line is written in ALL CAPS.
- sender_score decreases the likelihood of an observation being classified as spam. Here too, the score value is higher when both binary factors are equal to 1.

Predictions

As the last step of the analysis, let's take a look at the forecasts; they will be carried out on the dataset we previously separated from the training. In models like this, the predictions generate probabilities ranging from 0 to 1; once such values are found, through the assignment of an optimal threshold, the observations are grouped into the two classes in question (0 or 1).

```
##    is_spam Probabilities Posteriors
## 2        0  0.0060155552          0
## 3        0  0.0003292130          0
## 4        0  0.0002572593          0
## 5        1  0.9725944841          1
## 7        0  0.0075252825          0
## 14       1  0.4309960455          1
## 21       0  0.0010027659          0
## 24       0  0.0155356450          0
## 26       0  0.1463466780          0
## 27       0  0.0017234341          0
## 38       0  0.0141801563          0
## 40       0  0.1017083947          0
```

```
## 50       0  0.6230732889          1
## 52       0  0.0022746715          0
## 54       0  0.0014702064          0
## 56       0  0.0098188124          0
## 57       0  0.0441718982          0
## 61       0  0.0002606245          0
## 68       0  0.3595541106          1
## 73       0  0.0021712564          0
```

In this case, the optimal treshold is a value of approximately 0.34. By looking at the first 20 observations, we see that 18 are well classified, which may represent a great forecast.

To get more information about this, we can represent a confusion matrix, which shows information related to predictive accuracy, sensitivity, specificity...

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##          0 3518  119
##          1   91  272
##
##                Accuracy : 0.9475
##                  95% CI : (0.9401, 0.9542)
##     No Information Rate : 0.9022
##     P-Value [Acc > NIR] : < 2e-16
##
##                   Kappa : 0.6925
##
##  Mcnemar's Test P-Value : 0.06244
##
##             Sensitivity : 0.69565
##             Specificity : 0.97479
##          Pos Pred Value : 0.74931
##          Neg Pred Value : 0.96728
##              Prevalence : 0.09775
##          Detection Rate : 0.06800
##    Detection Prevalence : 0.09075
##       Balanced Accuracy : 0.83522
##
##        'Positive' Class : 1
##
```

From the table, we immediately notice that out of 4000 observations, about 3800 have been correctly assigned. In fact, a predictive accuracy of about 95% is recorded. Even specificity, which measures the model's ability to correctly identify instances that are truly negative, has a value very high.

3. Conclusions

In conclusion, the analysis highlighted the main determining factors for marking e-mails as spam, emphasizing how all the variables within the model, particularly the one containing the word 'offer', significantly affected the outcome. The use of logistic models and the analysis of marginal effects provided a detailed analysis

of the relationships between the independent variables and the response variable, highlighting how even the number of links present in the emails, the sender's score and the subject line written in ALL CAPS affect the probability that the mail will be classified as spam.

A crucial element for the classification was the choice of the optimal threshold. In fact, the analysis of accuracy, precision and recall metrics provided relevant insights to obtaining the predictions as accurately as possible.

Results that, from a practical usage perspective, are very important for individuals who work in jobs where there's frequent use of digital platforms, in order to avoid sudden phishing attacks; or even for people who are not very familiar with technology, such as our grandparents or parents (boomer generation).

4. Appendix

Model selection

Let's take a look to the processes that that led us to the selection of the optimal model.

In addition to the chosen model, the full model with 'logit' link, the full model with 'probit' and the full model with 'cloglog' link activation have been trained.

```
##
## Call:
## glm(formula = is_spam ~ ., family = binomial(link = "logit"),
##     data = train_s)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -6.1756800  0.2034294 -30.358   <2e-16 ***
## num_links     1.3461455  0.0366382  36.742   <2e-16 ***
## num_words     0.0003633  0.0007185   0.506    0.613
## has_offer1    4.2880853  0.1133073  37.845   <2e-16 ***
## sender_score -3.0343262  0.2038157 -14.888   <2e-16 ***
## all_caps1     3.1492691  0.1091041  28.865   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9828.5  on 15999  degrees of freedom
## Residual deviance: 4847.4  on 15994  degrees of freedom
## AIC: 4859.4
##
## Number of Fisher Scoring iterations: 7
```

The complete model shows that the variable num_words is not significant. This demonstrates that the number of words within the email does not appear to be influential for the analysis.

Since the two are nested models, let's compare them using the LRT (Likelihood-ratio test.)

```
## Analysis of Deviance Table
##
## Model 1: is_spam ~ num_links + num_words + has_offer + sender_score +
##     all_caps
```

```
## Model 2: is_spam ~ (num_links + num_words + has_offer + sender_score +
##     all_caps) - num_words
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1     15994     4847.4
## 2     15995     4847.7 -1 -0.25565   0.6131
```

The p-value strongly greater than the set threshold (0.05) suggests not to reject the null hypothesis that the two models are equal; therefore, considering the principle of parsimony (the less, the better), the choice falls on the reduced model.

Let's check the model with link function = 'probit':

```
##
## Call:
## glm(formula = is_spam ~ ., family = binomial(link = "probit"),
##     data = train_s)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.1782813  0.1030653 -30.838   <2e-16 ***
## num_links     0.7034243  0.0185425  37.936   <2e-16 ***
## num_words     0.0002273  0.0003841   0.592    0.554
## has_offer1    2.2011648  0.0558088  39.441   <2e-16 ***
## sender_score -1.6876297  0.1083508 -15.576   <2e-16 ***
## all_caps1     1.6328073  0.0568176  28.738   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9828.5  on 15999  degrees of freedom
## Residual deviance: 4883.8  on 15994  degrees of freedom
## AIC: 4895.8
##
## Number of Fisher Scoring iterations: 8
```

Here too there is a need to reduce the model, as once again the number of words is not statistically significant.

```
##
## Call:
## glm(formula = is_spam ~ . - num_words, family = binomial(link = "probit"),
##     data = train_s)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.15306    0.09391  -33.58   <2e-16 ***
## num_links     0.70345    0.01854   37.94   <2e-16 ***
## has_offer1    2.20083    0.05581   39.44   <2e-16 ***
## sender_score -1.68765    0.10835  -15.58   <2e-16 ***
## all_caps1     1.63264    0.05682   28.73   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
##     Null deviance: 9828.5  on 15999  degrees of freedom
## Residual deviance: 4884.1  on 15995  degrees of freedom
## AIC: 4894.1
##
## Number of Fisher Scoring iterations: 8
```

```
## Analysis of Deviance Table
##
## Model 1: is_spam ~ num_links + num_words + has_offer + sender_score +
##     all_caps
## Model 2: is_spam ~ (num_links + num_words + has_offer + sender_score +
##     all_caps) - num_words
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1     15994     4883.8
## 2     15995     4884.1 -1 -0.34663    0.556
```

Here too it's advisable to choose the reduced model.

Finally, let us examine the model using the complementary log-log link function.

```
##
## Call:
## glm(formula = is_spam ~ ., family = binomial(link = "cloglog"),
##     data = train_s)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -5.1582301  0.1541053 -33.472   <2e-16 ***
## num_links     0.9189508  0.0241452  38.059   <2e-16 ***
## num_words     0.0006288  0.0005519   1.139    0.255
## has_offer1    3.3129910  0.0822117  40.298   <2e-16 ***
## sender_score -2.1621530  0.1538100 -14.057   <2e-16 ***
## all_caps1     2.1146311  0.0743071  28.458   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9828.5  on 15999  degrees of freedom
## Residual deviance: 5041.7  on 15994  degrees of freedom
## AIC: 5053.7
##
## Number of Fisher Scoring iterations: 15
```

```
##
## Call:
## glm(formula = is_spam ~ . - num_words, family = binomial(link = "cloglog"),
##     data = train_s)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -5.08196    0.13985  -36.34   <2e-16 ***
## num_links     0.91769    0.02412   38.04   <2e-16 ***
```

```
## has_offer1    3.31207    0.08221   40.29   <2e-16 ***
## sender_score -2.16690    0.15378  -14.09   <2e-16 ***
## all_caps1     2.11274    0.07429   28.44   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9828.5  on 15999  degrees of freedom
## Residual deviance: 5043.0  on 15995  degrees of freedom
## AIC: 5053
##
## Number of Fisher Scoring iterations: 15
```

Now let's compare the three reduced models with the AIC and BIC information criteria.

```
##           df      AIC      BIC
## log_reg2   5 4857.655 4896.056
## prob_reg2  5 4894.102 4932.503
## clog_reg2  5 5052.984 5091.385
```

The values of AIC and BIC for the model with 'logit' link are lower, so it's better to use this kind of model.

Optimal threshold

```
##      threshold precision    recall        f1
## 3364 0.3378232 0.6956522 0.7493113 0.7214854
```

```
## [1] 0.3378232
```

The threshold was determined by finding the right balance between F1-score, sensitivity, and specificity.

$$Sensitivity = \frac{TN}{TN + FP} \qquad Specificity = \frac{TP}{TP + FN}$$

$$F1 - Score = 2 * \frac{NPV * Specificity}{NPV + Specificity}$$

In this way, by using this threshold, we see how we obtain the least number of incorrectly classified observations (210). By moving the threshold, we can reduce or increase the two metrics, depending on what we are really interested in. In this case, we are interested in finding emails that are spam (the positives), therefore the case where the variable equals 1. A proper method would be to reduce the threshold; in this way specificity increases but sensitivity decreases, along with accuracy. By doing this, I will obtain more misclassified observations, but I will have a greater number of positives.